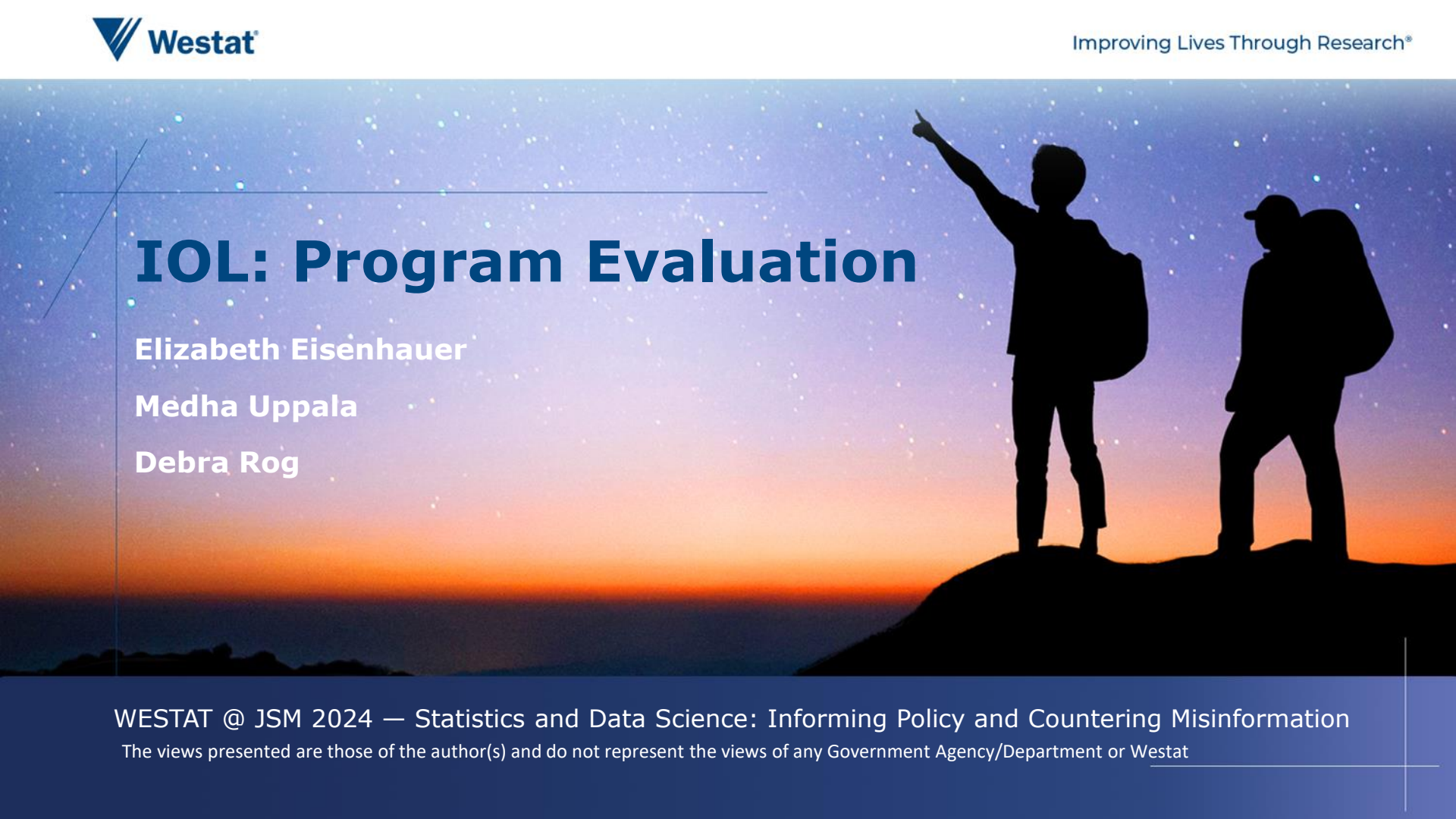




IOL: Program Evaluation

Elizabeth Eisenhauer

Medha Uppala

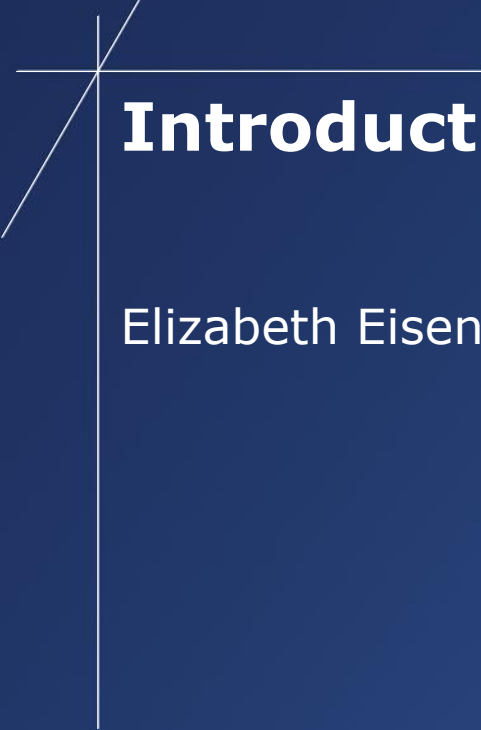
Debra Rog

WESTAT @ JSM 2024 — Statistics and Data Science: Informing Policy and Countering Misinformation

The views presented are those of the author(s) and do not represent the views of any Government Agency/Department or Westat

Agenda

- **Introduction to Program Evaluation** | Liz
- **Statistical Designs in Program Evaluation** | Medha
- **Combining Quantitative and Qualitative Approaches to Facilitate Evidence-based Decision Making** | Debra



Introduction to Program Evaluation

Elizabeth Eisenhauer

MY GOAL

To equip you with the tools to feel comfortable and motivated to **engage** with the program evaluation literature and community

What is a **program**?

A set of **resources and activities** directed toward one or more **common goals**, typically under the direction of a **manager or management team**.

A program announced in 2016 to train the U.S. workforce

- Funded by fees on H-1B visas, which allow employers to hire individuals from outside the U.S. to work in certain occupations
- TechHire and Strengthening Working Families Initiative (SWFI) train U.S. residents with barriers to skills training for H-1B industries

Program evaluation

- Department of Labor's Chief Evaluation Office and Employment and Training Administration
- Conducted by Westat and MDRC



What is **program evaluation**?

A **systematic** method for collecting, analyzing, and using data to examine the effectiveness and efficiency of programs and to contribute to continuous program improvement.

The program is also called the **evaluand** in this context.

Research vs. Evaluation

Research

Purpose: Testing theory and producing generalizable knowledge.

Evaluation

Purpose: Determine the effectiveness of a specific program or model.

Research vs. Evaluation

Research	Evaluation
Purpose: Testing theory and producing generalizable knowledge.	Purpose: Determine the effectiveness of a specific program or model.
Research questions come from scholars in a discipline.	Evaluation questions come from stakeholders and intended users of evaluation findings.

Research vs. Evaluation

Research	Evaluation
Purpose: Testing theory and producing generalizable knowledge.	Purpose: Determine the effectiveness of a specific program or model.
Research questions come from scholars in a discipline.	Evaluation questions come from stakeholders and intended users of evaluation findings.
Value: Contribution to knowledge base.	Value: Usefulness to improve program implementation and effectiveness.

Research seeks to prove, evaluation seeks to improve.

Michael Quinn Patton, Founder and Director,
Utilization-Focused Evaluation

Federal program evaluation standards

- Relevance and Utility
- Rigor
- Independence and Objectivity
- Transparency
- Ethics

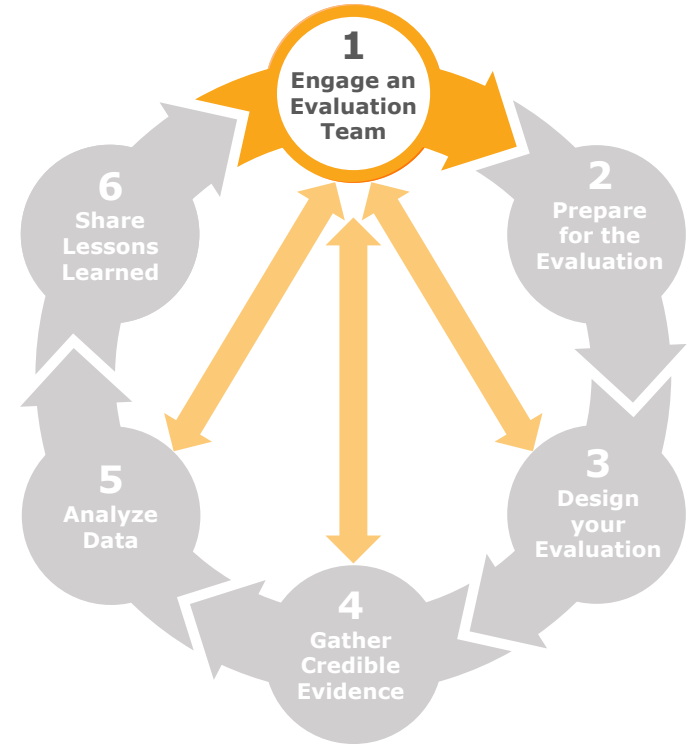
Basic steps in program evaluation



Iteratively engage an evaluation team

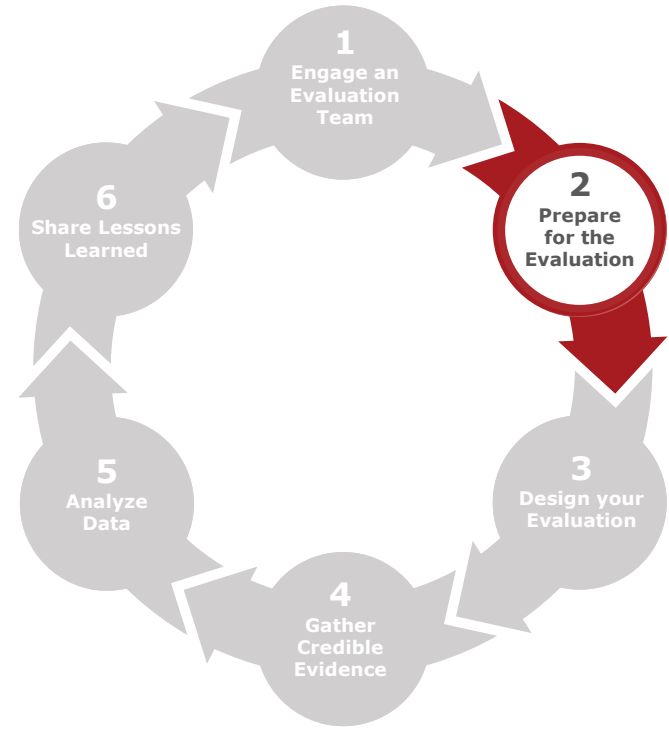
The evaluation team includes:

- **Principal Investigator**
- Subject matter **experts**, including statisticians
- Interested parties and intended users:
 - **Engaged** in operations
 - **Served or affected**
 - Intended **users** of the findings

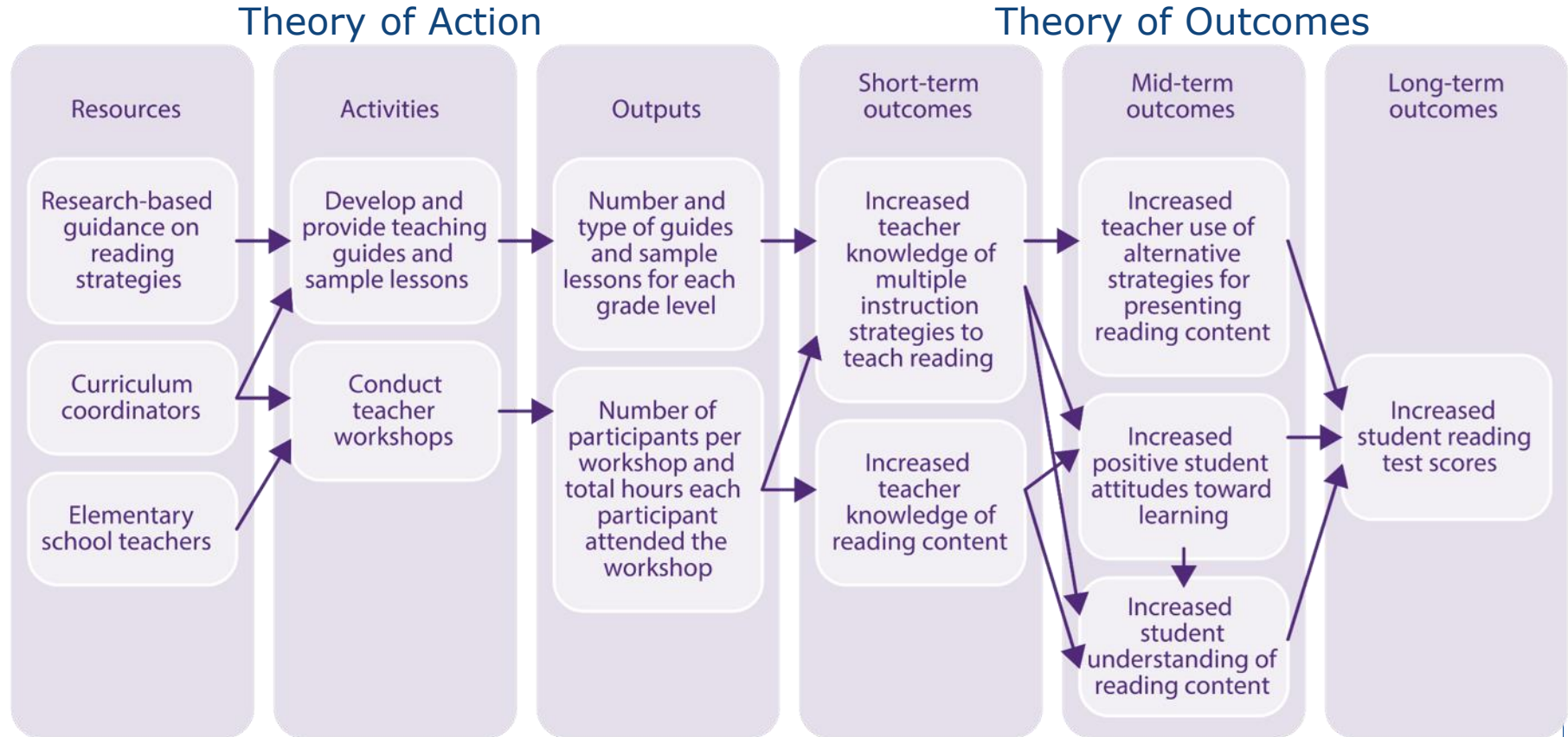


Define the evaluand, questions, and logic

- The **evaluand** can be the entire program, or a specific component, service, or activity.
- Evaluation questions cover:
 - Has the program been successful in attaining the anticipated **implementation & impact** objectives?
- Link these concepts in a **logic model**.



Logic models describe the program theory

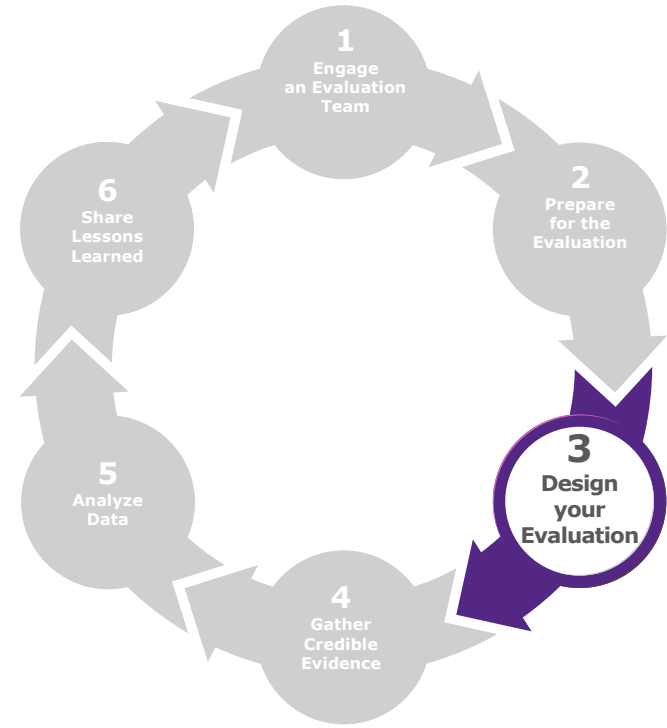


Cost drivers and cost savers

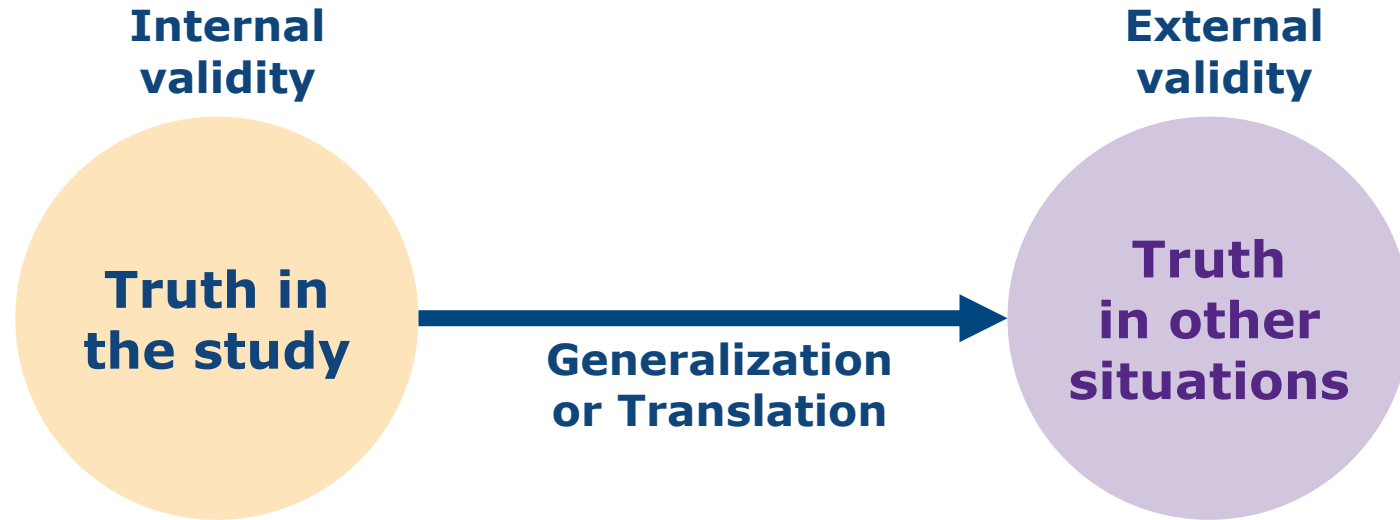
- What **data** are available (secondary data) and what could be collected (primary data)?
- What statistical designs might be needed to establish causality? **Stronger evidence of causality = higher cost**
- Does the program already have the **capacity** to conduct the evaluation? If not, what is needed?

Evaluation design and making it happen

- **Design** | approach to answer each evaluation question
 - Descriptive
 - Experimental
 - Quasi-experimental
 - Mixed methods
- **Data collection** | strategy to execute that plan
 - Surveys, interviews, administrative data, focus groups, observations, document review



2 types of validity in the evaluation design



Supports | random assignment, high response rates

Threats | crossover (comparison groups getting the treatment)

Supports | implement in the “real world” context

Threats | infeasible supports or incentives

What makes evidence “credible”?

- For example,
 - If **family financial outcomes** are important,
 - One might measure **whether parents own a vehicle**
 - Participants say, “**we use public transportation so why get a car?**”
 - Measuring only car ownership is not **relevant**, or will not generate evidence that is of interest, or credible, to this community



What makes an evaluator “credible”?

Accidental Evaluators

- Many evaluators “fall into” evaluation, often without formal education in evaluation

Reflexivity/Positionality

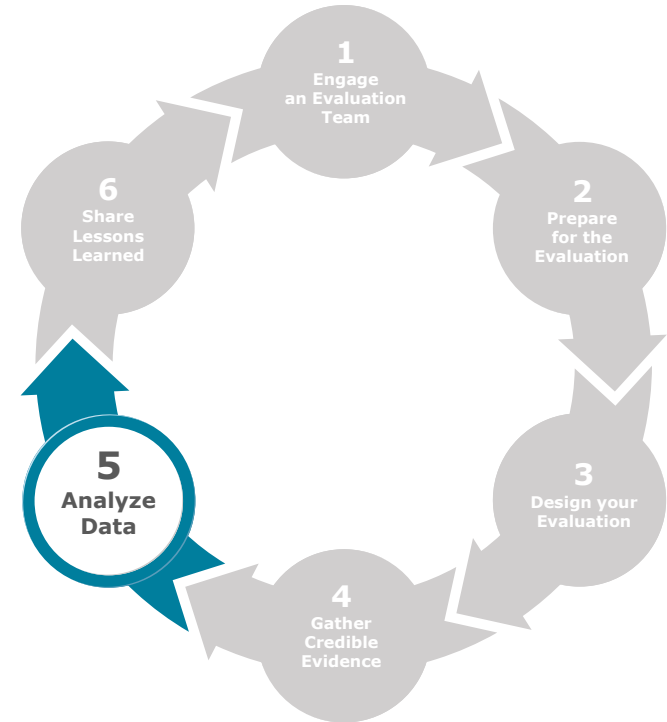
- Development of cultural humility by questioning and exploring your underlying values, assumptions, and beliefs that influence the evaluation process

AEA Evaluator Competencies

- Professional practice
- Methodology
- Context
- Planning & management
- Interpersonal

Analyses are ongoing and engage the evaluation team

- Ongoing analyses
 - Help facilitate **midcourse corrections** to improve data quality
 - **Identify challenges** in program implementation
 - **Inform expectations**
- Engage the team to
 - Maintain the **dignity** of participants
 - Provide **context**
 - **Disaggregate** data to illuminate subgroup differences



Sharing findings—good, bad, or null—ensures transparency and can foster positive change.

Communication **goals** include:

- **Advancing the field, improving programs and policies, and sensemaking**
- Getting **attention from potential funders** to support the program or further evaluation
- Increasing **awareness** of the program



A program announced in 2016 to train the U.S. workforce

- Funded by fees on H-1B visas, which allow employers to hire individuals from outside the U.S. to work in certain occupations
- TechHire and Strengthening Working Families Initiative (SWFI) train U.S. residents with barriers to skills training for H-1B industries

Program evaluation

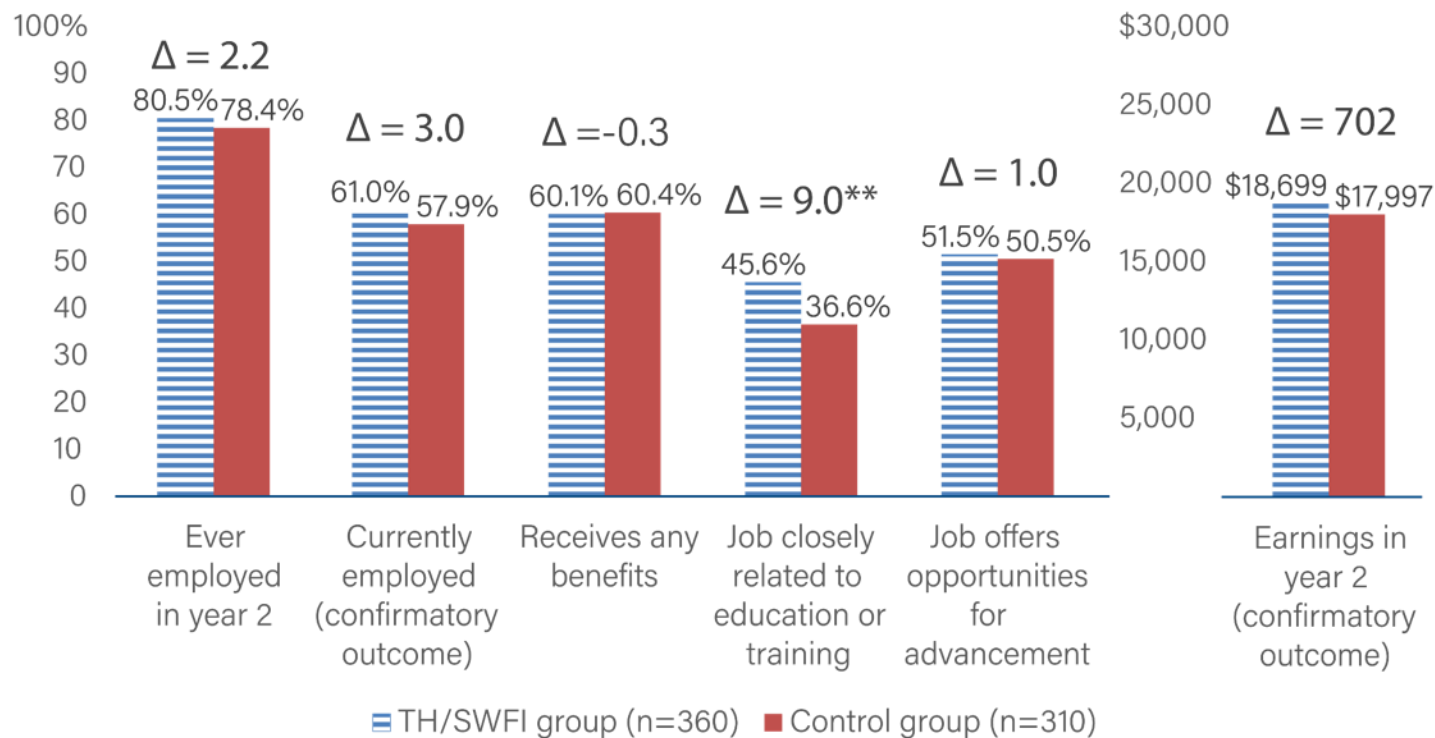
- Department of Labor's Chief Evaluation Office and Employment and Training Administration
- Conducted by Westat and MDRC



This program evaluation used an experimental design

- 5 sites conducted randomized controlled trials in 2018-2020
 - **TechHire/SWFI group** was eligible to receive TechHire- or SWFI-funded program training and services
 - **Control group** was not eligible for these services but could receive other training or services from other sources
- 1-2 years later, **43%** of TechHire/SWFI participants had completed training compared to **21%** of control participants
- Key outcomes:
 - Employment
 - Earnings

Impact on training completion did not translate into impacts on employment and earnings outcomes



Implementation lessons from a randomized controlled trial

- **Program recruitment** was a challenge.
- Challenges preparing participants with low skills to enter **high-skilled, high-tech jobs in a relatively short period of time.**
- Of the support services that programs offered, **case management was widely valued, while childcare services were infrequently used.**
- **Relationships with employers were key**

How can I become involved as a statistician?

- **Get to know people doing evaluations**

- Develop those connections

- **Help design the study**

- Examine tradeoffs for different study designs (e.g., options for random assignment)
- Think about the measurement and analysis along with the design

- **Engage as early as possible and stay involved**

- Seeing different perspectives such as qualitative analyses will help you build a better study

- **Don't fall into the trap of evaluating the things that are easy to evaluate. Remember what is important within the context.**

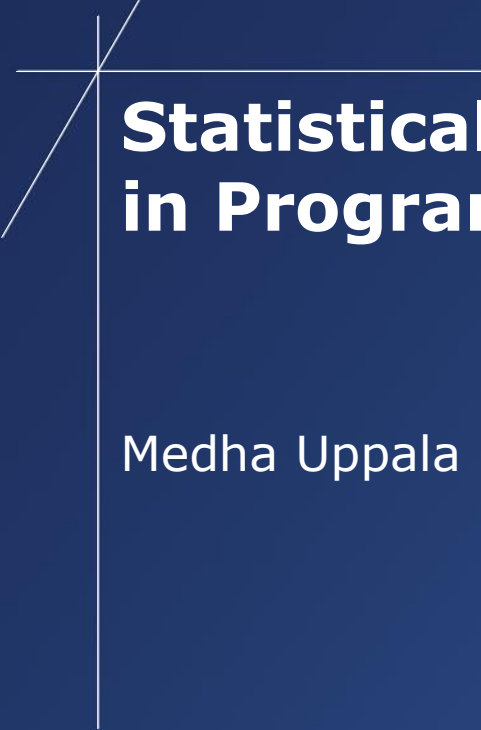
Related, upcoming JSM sessions



- **Program and Policy Evaluation:
The Role of the Statistician (invited panel)**
 - Tues, Aug 6: 2:00 PM, CC-F150
- **The U.S. Federal Statistical System:
Why We're Here, What We Do, Who We Are,
and Career Opportunities (IOL)**
 - Wed, Aug 7: 8:30 AM, CC-251

References

- Attia, M., & Edge, J. (2017). Be(com)ing a reflexive researcher: A developmental approach to research methodology. *Open Review of Educational Research*, 4(1), 33–45. <http://dx.doi.org/10.1080/23265507.2017.1300068>
- El Mallah, S., Gutuskey, L., Hyra, A., Hare, A., Holzwart, R., & Steigelman, C. (2022). *The program manager's guide to evaluation* (OPRE Report 2022-208). U.S. Department of Health and Human Services, Administration for Children and Families, Office of Planning, Research, and Evaluation.
- Gasper, Joseph, Joshua Vermette, Kevin Baier, Kelsey Schaberg, and Richard Hendra. (2023). *Evaluation of Strategies Used in the TechHire and Strengthening Working Families Initiative Grant Programs: Two-Year Impacts Report*. Washington, DC: U.S. Department of Labor.
- Kekahio, W., Cicchinelli, L., Lawton, B., & Brandon, P. R. (2014). *Logic models: A tool for effective program planning, collaboration, and monitoring*. (REL 2014–025). Washington, DC: U.S. Department of Education, Institute of Education Sciences, National Center for Education Evaluation and Regional Assistance, Regional Educational Laboratory Pacific. Retrieved from <http://ies.ed.gov/ncee/edlabs>.
- King, J. A., & Stevahn, L. (2020). Presenting the 2018 AEA Evaluator Competencies. In J. A. King (Ed.), *The American Evaluation Association's Program Evaluator Competencies*. *New Directions for Evaluation*, 2020, 49–61.
- LaVelle, J. M., Jones, N. D., & Donaldson, S. I. (2024). Addressing the Elephant in the Room: Exploring the Impostor Phenomenon in Evaluation. *American Journal of Evaluation*, 45(2), 186-202. <https://doi.org/10.1177/10982140221075243>
- Wanzer, Dana Linnell. "What is evaluation?: Perspectives of how evaluation differs (or not) from research." *American Journal of Evaluation* 42.1 (2021): 28-46.
- Massengale, K. E. C., Comer, L. H., Austin, A. E., & Goldblum, J. S. (2020). [Diaper need met among low-income US children younger than 4 years in 2016](#). *American Journal of Public Health*, 110, 106-108.
- Newcomer, K. E., Hatry, H. P., & Wholey, J. S. (Eds.). (2015). *Handbook of Practical Program Evaluation* (4th ed.). Jossey-Bass.
- Patton, Michael Quinn (2014). *Evaluation Flash Cards: Embedding Evaluative Thinking in Organizational Culture*. St. Paul, MN: Otto Bremer Foundation, ottobremer.org. https://ottobremer.org/news_stories/evaluation-flash-cards/
- Rock, Andrew, Rachael Metz, Betsy Tessler, and Joseph Gasper. 2021. *Implementation Considerations for Practitioners from the TechHire and SWFI Randomized Controlled Trial*. Brief prepared for the U.S. Department of Labor, Chief Evaluation Office. Rockville, MD: Westat; and New York: MDRC.
- Wanzer, D. L. (2021). What Is Evaluation?: Perspectives of How Evaluation Differs (or Not) From Research. *American Journal of Evaluation*, 42(1), 28-46. <https://doi.org/10.1177/1098214020920710>



Statistical Designs in Program Evaluation

Medha Uppala

Overview of Quantitative Methods in Evaluation

- Descriptive Designs
 - "Not meant to study cause and effect relationships"
 - Meant to get a picture of the phenomenon
 - Answers questions such as 'what is' or 'how much' etc.
- Primer on Causal Inference
- Quasi-Experimental Designs

Primer on Causal Inference

- Motivated by the question of 'cause and effect'
 - Does X cause Y?
 - If so, how large is the effect of X on Y? Is the effect size large relative to other causes of Y?
- 2 Frameworks or Approaches
 - Potential Outcomes Framework
 - Directed Graph Framework

Primer on Causal Inference: Potential Outcomes Model

- A core component in the counterfactual framework
- Origins in early work on experimental design by Neyman (1923, 1935), Fisher (1935), Cochran and Cox (1950), Kempthorne (1952), Cox (1958).
- Formalized in a series of papers by Rubin (1974, 1977, 1978, 1980a, 1981, 1986, 1990).
- Term "potential outcome" is a reference to Neyman's work on potential yields in agricultural statistics.

Primer on Causal Inference: Potential Outcomes Model

- Population of interest
- Exposed to multiple alternative causal states
 - Binary case: Treatment and Control
- Causal exposure variable (treatment), $D \in \{0,1\}$
- Potential outcome random variables for each unit i , Y_i^1 and Y_i^0 .
- Realized versions are d_i , y_i^1 and y_i^0 .

Primer on Causal Inference: Potential Outcomes Model

- Fundamental problem of causal inference (Holland 1986)
 - Impossible to calculate individual level causal effects

Group	Y^1	Y^0
Treatment (D = 1)	Observable as Y	Potential Outcome*
Control (D = 0)	Potential Outcome	Observable as Y

- Individual treatment effect for unit i ,

$$Y_i^1 - Y_i^0$$

- (Unconditional) Average treatment effect (ATE):

$$E[\delta] = E[Y^1] - E[Y^0]$$

*Potential outcomes under D and Counterfactual under d (treatment assignment has occurred) Rubin (2005)). Common for social scientists to use the term counterfactual as treatment *selection* already occurs in an observational setting.

Primer on Causal Inference: Potential Outcomes Model

Group	Y^1	Y^0
Treatment ($D = 1$)	Observable as Y $E[Y^1 D = 1]$	Potential Outcome $E[Y^0 D = 1]$
Control ($D = 0$)	Potential Outcome $E[Y^1 D = 0]$	Observable as Y $E[Y^0 D = 0]$

- (Conditional) Average Treatment effect for the treated (ATT):

$$E[\delta | D = 1] = E[Y^1|D = 1] - E[Y^0|D = 1]$$

- (Conditional) Average Treatment effect for the control (ATC):

$$E[\delta | D = 0] = E[Y^1|D = 0] - E[Y^0|D = 0]$$

Primer on Causal Inference: Potential Outcomes Model

- **Stable Unit Treatment Value Assumption (SUTVA):** Outcome value Y of a unit is the same
 - i. no matter the treatment mechanism* used to assign treatment D
 - ii. no matter the treatment the rest of the units receive
- In other words, outcome is independent (not causally affected) of the treatment assignment mechanism *and* the treatment vector $\mathbf{d}$

$$y_i^1(\mathbf{d}) = y_i^1 \text{ and } y_i^0(\mathbf{d}) = y_i^0$$

- E.g. Study participant's health outcome to a drug administered to another participant vs. Study participant's infection rate to a vaccine administered to a household member

*No unrepresented versions of treatments

Primer on Causal Inference: Potential Outcomes Model

- **Ignorability**

Unit's receipt of treatment is independent of its potential outcomes
RCTs fulfil this assumption

$$(Y^0, Y^1) \perp D$$

- In other words, there is no selection bias or self-selection into a treatment group
- Generally difficult to fulfil in social sciences and observational studies

Quasi-Experimental Designs

- When pure experiments are not possible, the fall-back methods are 'Quasi (as if)-experimental'. Emulate experimental setup as much as possible
- Origins in the fields of Sociology, Demography and Psychology
 - RCTs aren't easy to perform
 - Use of experimental language in observational social sciences
- Psychologist Campbell and colleagues introduce the term of Quasi-experiments.
- Quasi-experiments "have treatments, outcome measures, and experimental units, but do not use random assignment to create the comparisons from which treatment-caused change is inferred" Cook and Campbell (1979:6)

Quasi-Experimental Designs

- These designs took off in program evaluation but lost steam in core social statistics
- Some Quasi-experimental designs include:
 1. (One-group) pretest-posttest design
 2. Nonequivalent group designs
 3. Multiple measures: Interrupted Time-Series designs
 4. Regression Discontinuity designs

(One-group) Pretest-Posttest Designs

- Comparison of what happened before the treatment to what happened after the treatment in the same set of units.

Measurement	Treatment	Measurement
O	X	O

- As the same set of participants are observed pre and post test, all their potential outcomes are observable:

Group	y^0	y^1
One group $D = [0, 1]$	Observable	Observable (at a different time point)

(One-group) Pretest-Posttest Designs continued

- Easy to implement
- Threats to internal validity:
 - History | Alternative confounders before treatment
 - Maturation | Alternative temporal effects?
 - Instrumentation | measurement bias due to changes in definitions or instruments over time.
 - Regression towards to the mean, “spontaneous remission”
 - Participant Attrition
- Link to Interrupted time series designs

Nonequivalent group Designs

- Estimate treatment effect by comparing different groups.
 - Due to absence of treatment randomization, the groups are 'non-equivalent'.
 - Posttest-only nonequivalent group design:

	Treatment	Measurement
	X	O
		O

Group	Y^0	Y^1
Treatment (D=1)	Unobserved	Observable
Nonequivalent Control (D=0)	Observable	Unobserved

- Threat to internal validity: Selection Bias – Y is not independent of D

Nonequivalent group Designs | continued

- Pretest-posttest nonequivalent group design:

Measurement	Treatment	Measurement
O_T	X	O_T
O_C		O_C

Group	Y^0	Y^1
Treatment ($D=1$)	Observable	Observable (at a different time point from Y^0)
Nonequivalent Control ($D=0$)	Observable	Unobservable

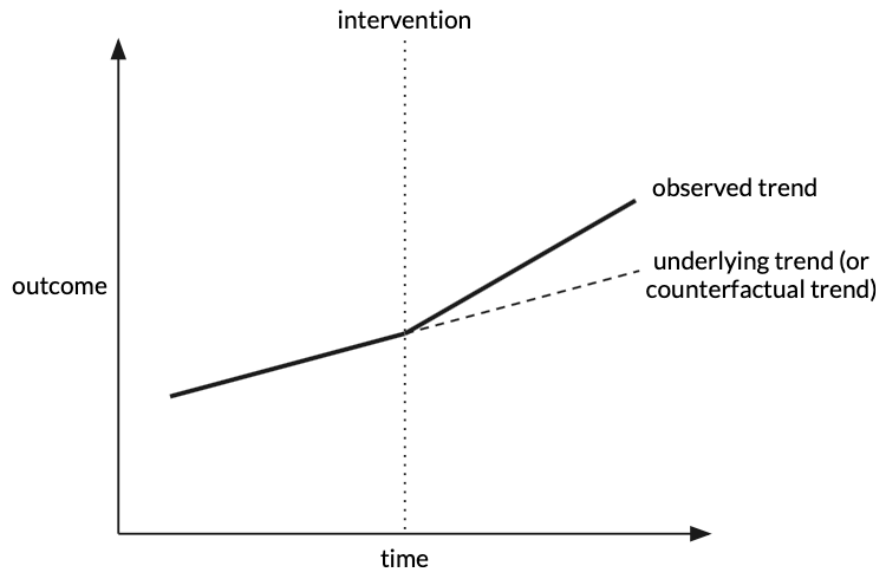
- Threats to internal validity for pretest-posttest nonequivalent group design:
 - Selection bias: Selection interacting with confounders, e.g. selection-by-maturation
- Solutions include:
 - Analysis of Covariance (ANCOVA)
 - Latent variable structural equation models
 - Matching methods like Propensity Score analysis
- Link to Difference in Differences

Interrupted Time-Series Designs

- A time-series of observations interrupted by a treatment
- One-group interrupted time-series design:

○ ○ ○ ○ ○ ○ X ○ ○ ○ ○ ○ ○

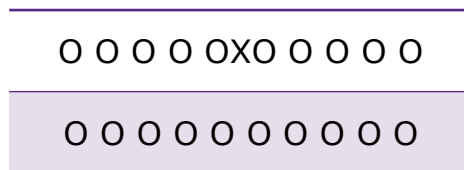
- Trend in the pre-treatment outcome is estimated and projected forward to provide a no-treatment counterfactual.
- Observed, post-treatment outcome trend is compared to the estimated counterfactual.



- Example: Measure of worker productivity every week in a year, with treatment being reduction in length of work shift from 10 to 8 hours.
- Threats to interval validity:
 - Immediate or delayed effects, history and maturation's effects, temporary or permanent effects.
 - Disentangling trend and treatment effects
 - Instrument bias, attrition and length of time interval are limiting factors.

Interrupted Time-Series Designs | continued

- Interrupted time series design with control:



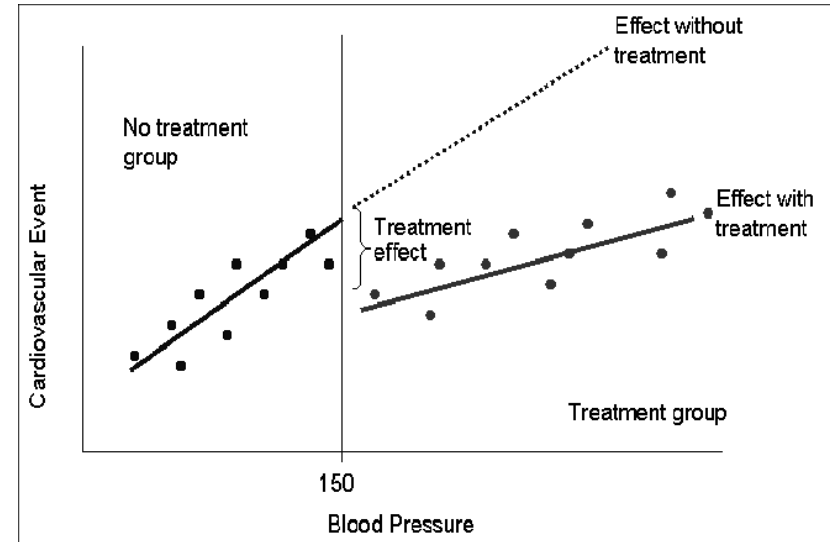
- Better information on the counterfactual if the groups are equivalent
 - Differences-in-Differences estimators
- Interrupted time-series with removed and repeated treatment:



- ARIMA, Hierarchical Linear Modeling, Growth curve modeling.

Regression-Discontinuity Design

- Similar to interrupted time-series designs except that the treatment assignment is a function of values of a certain variable rather than time.
- Participants are assigned to treatment groups based on their scores on a continuous measure
 - Such a measure is called Quantitative Assignment Variable
- Similar challenges as ITS designs



Ties to Causal inference Methods

- **SUTVA violation**

- Estimating Spillover effects that account for interference

- **Ignorability (Selection Bias)**

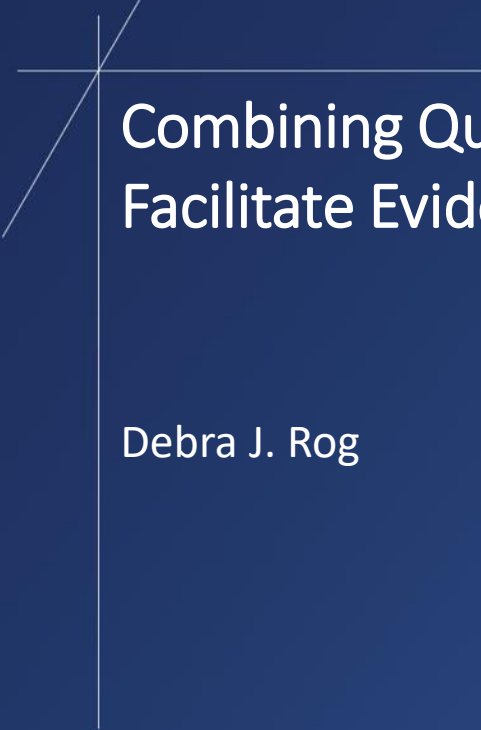
- Stratification, matching method (Propensity Scores and IPW)
- Regression estimators, weighted, doubly robust estimators
- Time-series modeling

- **Internal validity**

- Identifying confounders and colliders
- Use of mediators, moderators or instrumental variables

References

- Morgan SL, Winship C. Counterfactuals and Causal Inference: Methods and Principles for Social Research. 2nd ed. Cambridge, UK: Cambridge University Press; 2014.
- Bickman L, Rog DJ. The Sage Handbook of Applied Social Research Methods. 2nd ed. Sage Publications Inc; 2008.



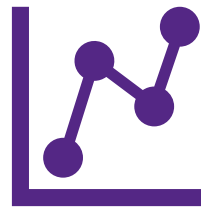
Combining Quantitative and Qualitative Approaches to Facilitate Evidence-based Decision Making

Debra J. Rog

Introduction to “Mixed Methods” Designs



**The what and why
of mixed methods
designs**

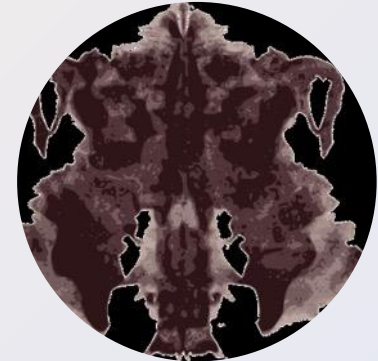


Overview of:

- Designs
- Sampling and data collection
- Analytic strategies
- Drawing inferences

What is a Mixed Methods Design?

Collection, analysis,
integration and interpretation
of qualitative and quantitative
data in a single study, guided
by research questions and
theoretical framework



Research Questions Drive the Design

Mix of how and why questions with what questions

- What is the effect of program x on participants?
- Why are some participants more receptive to the program than others?
- How does the nature of the services provided affect how receptive participants are?

Questions can be parallel or sequential

- What proportion of participants use hospitals for primary care?
- Why do participants use hospitals? Why does use differ by different subpopulations?
- What alternatives do they have? What are the barriers to using these alternatives?

Key Words to Walk Away With

Integration



Intentionality



Value-Add of Mixed-Methods



PRACTICAL; OFTEN
NEEDED TO TELL
THE WHOLE STORY



CAN COMBINE
DIFFERENT WORLD-
VIEWS IN A SINGLE
EVALUATION



OFFERS BOTH
BREADTH AND
DEPTH



BRIDGES
DIFFERENT WAYS
OF KNOWING

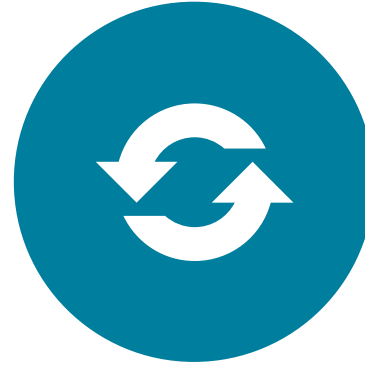


PROVIDES
GREATER
EVIDENCE FOR
DECISION-MAKING
AND
COMMUNICATION

Two Most Common Types of Designs



Parallel



Sequential

Adding Qualitative Methods to RCTs/QEs

Before

(Exploratory Sequential)

- **Designing** the study
- Assessing **feasibility**
- Developing **instruments**
- Understanding the **context**
- Documenting the **need** for the program
- Informing **recruitment**

During

(Parallel)

- **Validating** quantitative outcomes
- Understanding **impact beyond quantitative (facilitators, barriers)**
- Assessing **unanticipated** consequences
- **Understanding context**
- Understanding **participant experience**
- Assessing **fidelity**

After

(Explanatory Sequential)

- Obtaining **participants' views on results**
- Conduct studies to **explain** outcomes
- Determine **sustainability** of the program/intervention

Mixed-Methods Sampling



Combinations of quantitative sampling (e.g., random, convenience) and qualitative sampling (e.g., purposive samples such as “typical”, “extremes”)



Creative combination of techniques is a defining characteristic



Generate complementary databases that have both breadth and depth



Typically, multiple samples – e.g., state agency directors, provider directors, provider staff, population served

Aligning Mixed Methods Samples

Parallel

- Could be exactly the same, could have subsamples that overlap or could be completely different

Explanatory Sequential

- Followup qualitative sample usually drawn from the initial QUAN study, and typically a small number

Exploratory Sequential

- QUAL study prior to QUAN study and samples usually differ, generally due to pragmatics. QUAN is also typically much larger.

Evaluation of Public Health Law Program

Quantitative | Surveys

- Random samples of state and local health officials
- Complete universe of researchers
- Random sample of individuals contacting the Network

Qualitative | Case Studies

- Tracer cases – selection of recommended cases meeting criteria of impact, snowball samples of individuals to interview
- Hot topic cases – where critical mass of attention in field and in PHL program; random sample of states meeting criteria of recent legislation, etc.; snowball sampling of individuals

Data Collection Framework

Research Questions or Domains	Desk Reviews	Interviews	Web based survey	Secondary data	Pop. Based survey	Observations
Question 1	X		X		X	
Question 2	X	X	X	X	X	
Question 3	X		X		X	
Question 4	X	X	X	X	X	X
Question 5	X	X	X	X	X	
Question 6	X		X	X		
Question 7	X		X			

Tools/Systems for Mixed-Methods Analysis

Conceptual frameworks

Logic models
Theories of change

Evaluation Plans

Overall plans

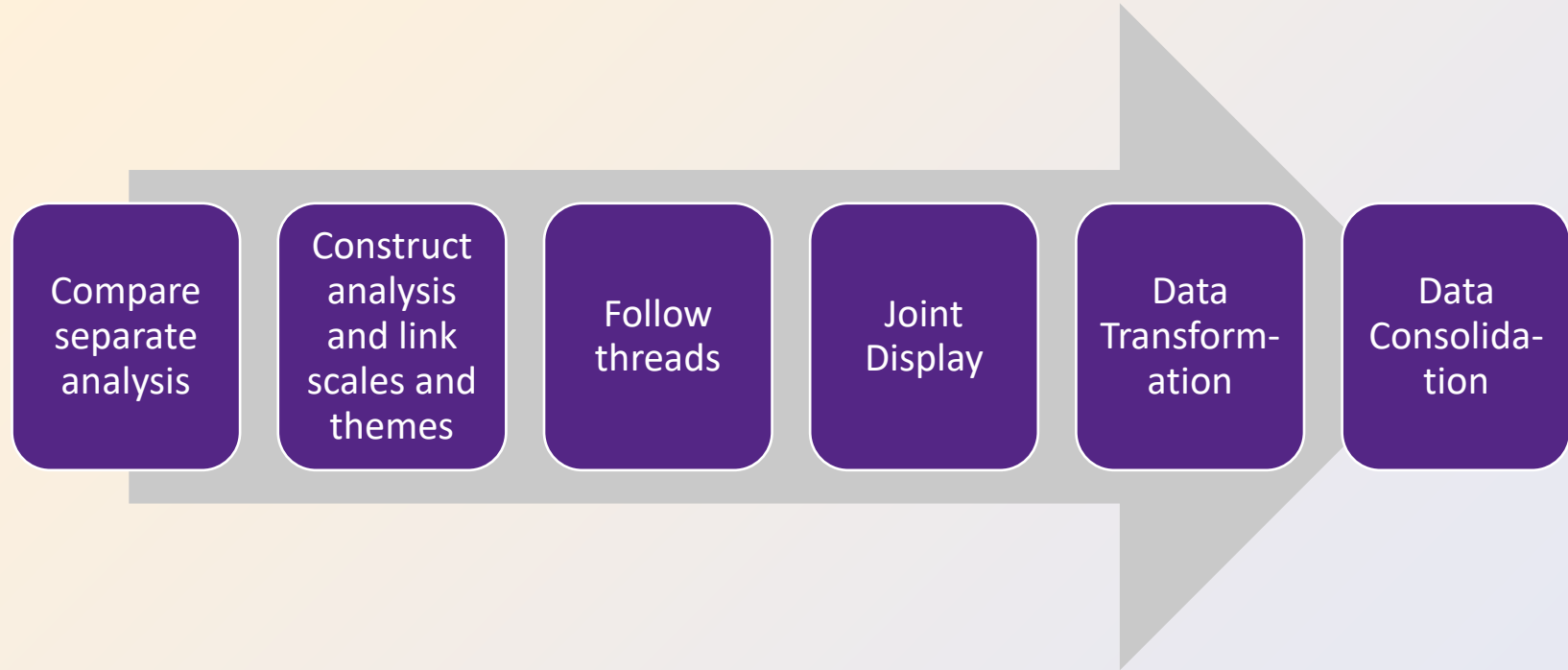
Data matrices and collection plans, analysis plans

Analysis Tools

Use of databases, organizing tools

Matrix, joint displays, data reduction strategies

Techniques for Merging QUAL and QUAN



Techniques for Merging QUAL and QUAN

NARRATIVE INTEGRATION

Compare
separate
analysis

Construct
analysis
and link
scales and
themes

Follow
threads

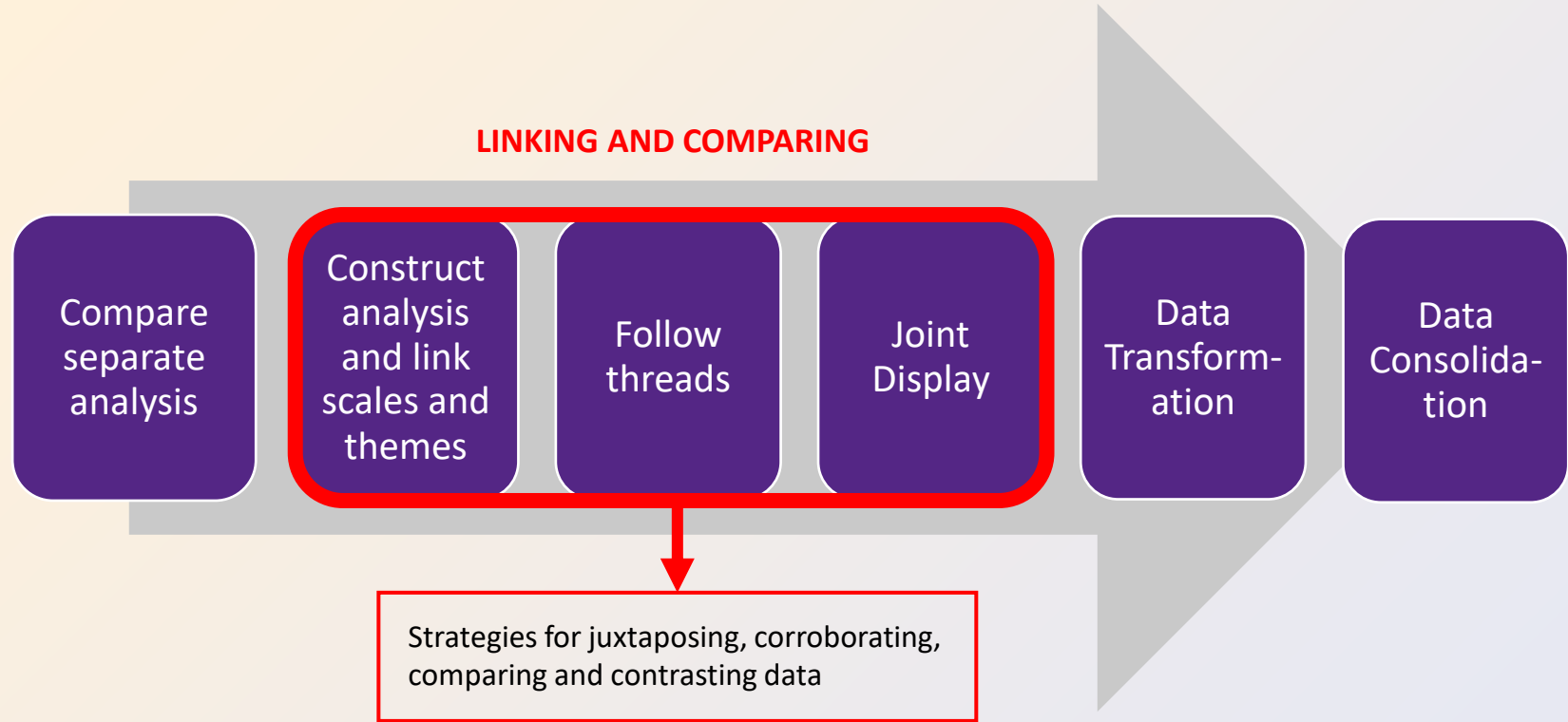
Joint
Display

Data
Transfor-
mation

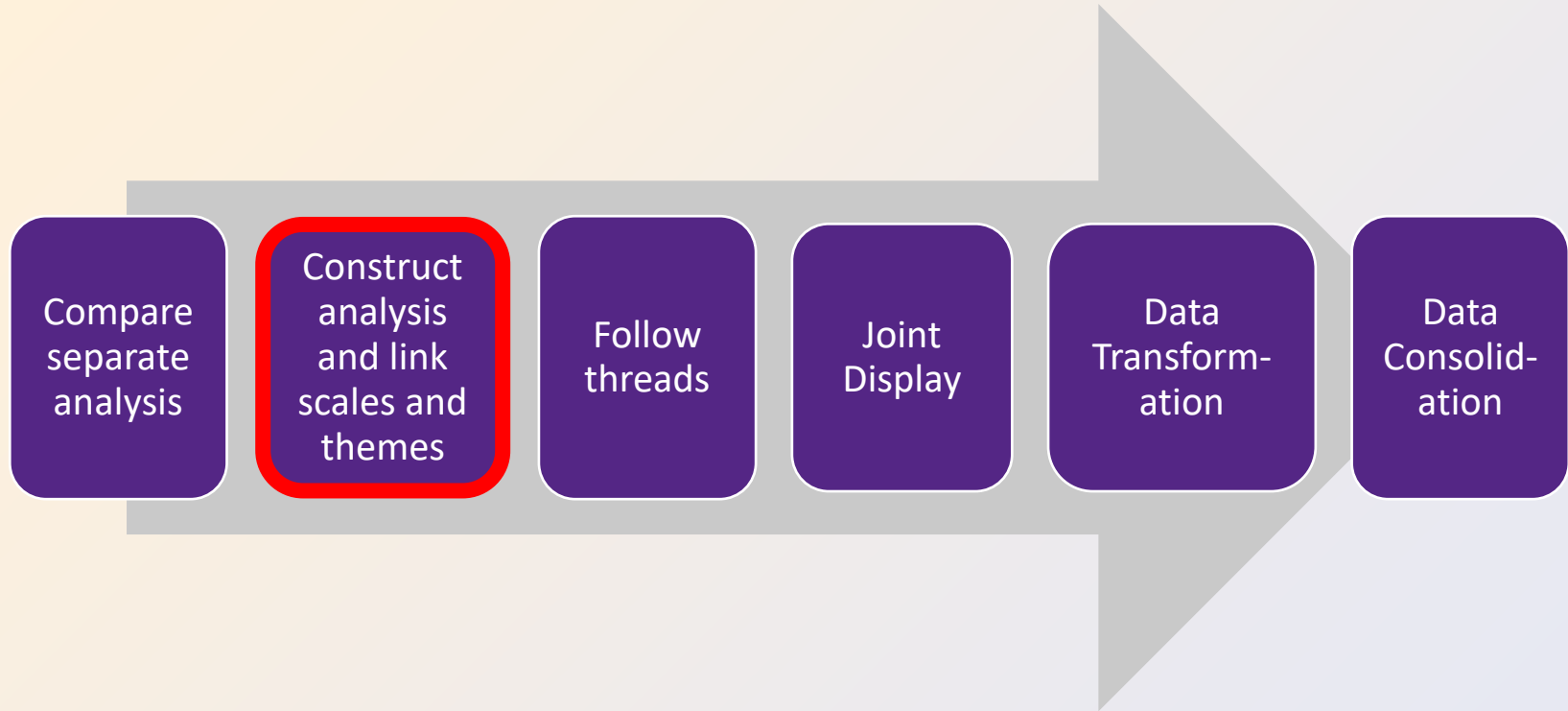
Data
Consolid-
ation

No integration at analysis; narrative results are compared, contrasted, and summarized.

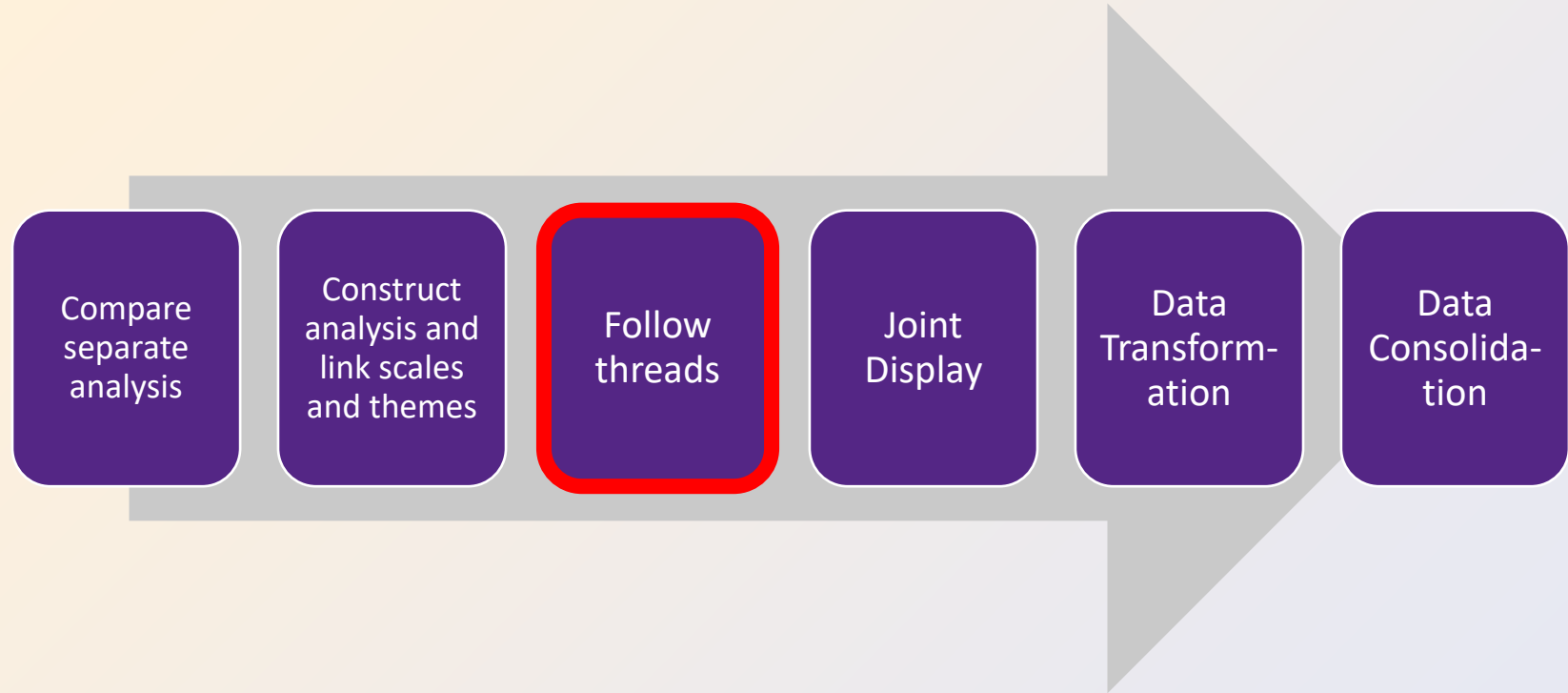
Techniques for Merging QUAL and QUAN



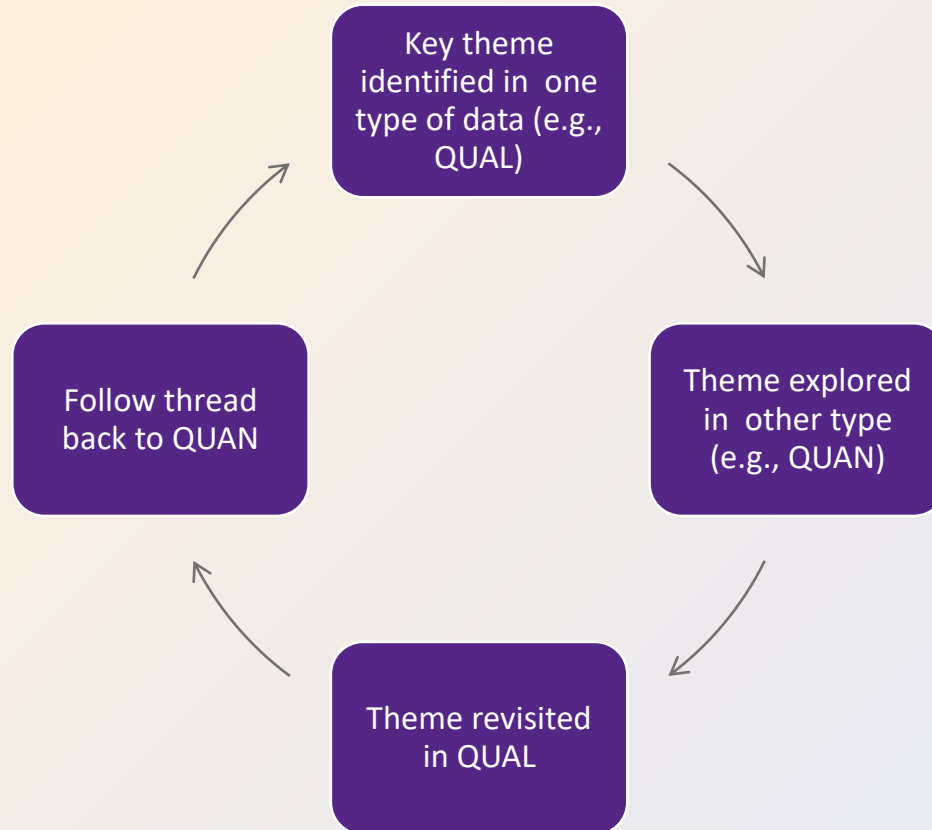
Techniques for Merging QUAL and QUAN



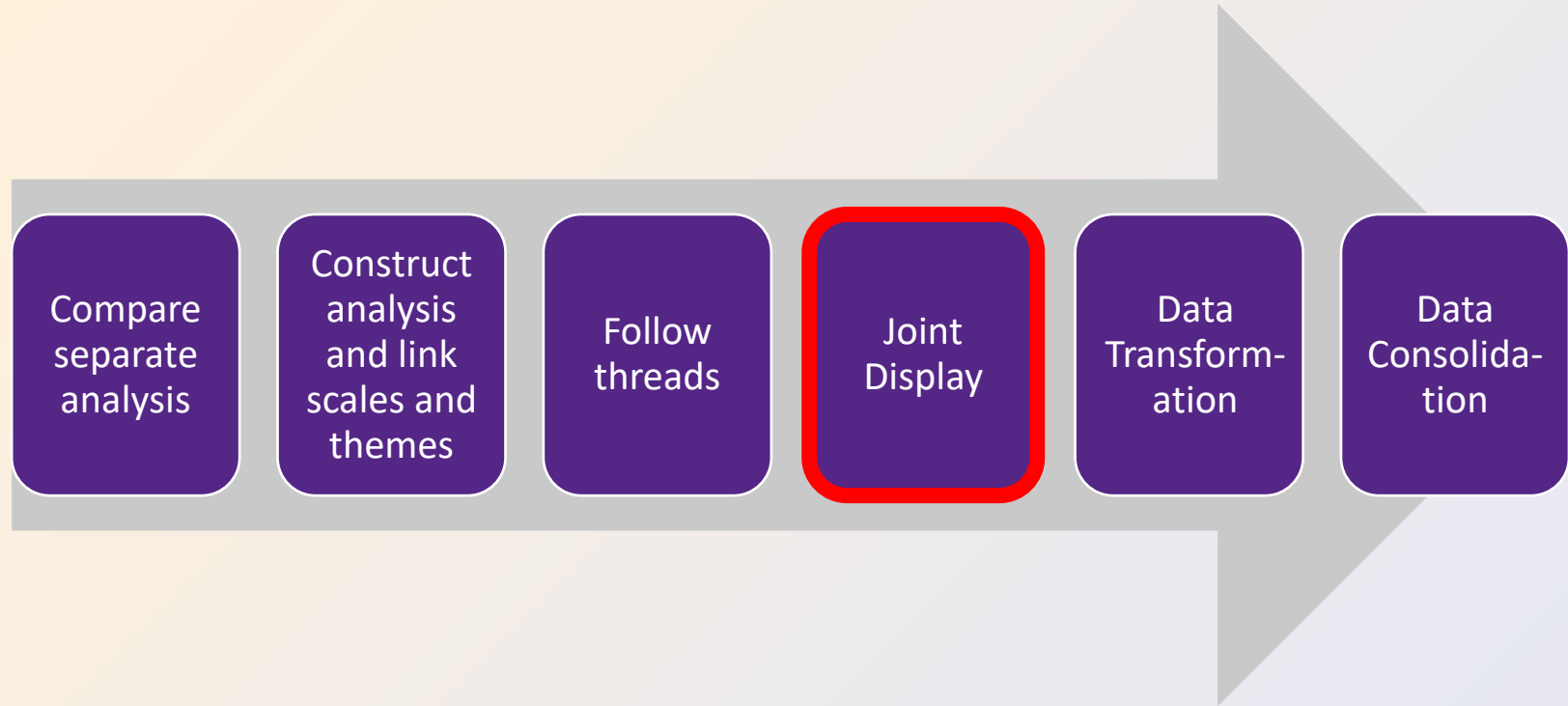
Techniques for Merging QUAL and QUAN



Example of Following A Thread



Techniques for Merging QUAL and QUAN



Types of Joint Displays (Creswell)

Type of Design	MM Questions	Merging/ Connected Data Analysis	Types of Joint Displays
Parallel Design	To what extent to QUAL and QUAN results converge?	Merging analysis	• Side by side • Combine themes/categories • Relate QUAL themes to QUAN ratings to inform QUAL transformation
Explanatory Design	In what ways do the QUAL data help to explain the QUANT results?	Building/ sequential	• Link qual themes to quant results to explain
Exploratory Design	In what ways do the QUAN results generalize the QUAL findings?	Building/ sequential	• Quotes, codes and themes that match items, variables, etc. • How QUAN results generalize QUAL themes

Example of a Joint Display

Table 4 Joint display of patient experiences per treatment benefits

Treatment benefits	Change in music therapy ^a	Change in music medicine ^a	Patient experiences
↑MT, ↓ MM	0.65 to 1.88	-0.11 to 0.38	• Emphasize the importance of therapeutic relationship and support by therapist • Enjoy the creative aspect of music making • Are hopeful for the future
↑MM, ↓ MT	-0.46 to 0.59	0.33 to 1.63	• Apprehensive about active music making • Prefer familiarity of pre-recorded music • Hesitant about exploring feelings related to cancer
↑MT, ↑ MM	0.61 to 1.07	0.73 to 1.37	• Strong conviction about the power of music to support and give hope • Use music for mental escape • Use music for emotional exploration and value processing of emotions with therapist
↓ MT, ↓ MM	-0.67 to -1.03	-0.52 to -1.06	• Hold little hope for the future • Music evokes sad and traumatic memories • Feel inadequate regarding music making and singing • Prefer aesthetics of original recordings

↑ great improvement, ↓ less improvement or worsening

^a Range of ~~mean~~ (average of z-scores for mood, anxiety, depression and pain)

Reprinted with permission from *Support Care Cancer*. 2015;23(5):1261-1271.²⁹

Figure

Caption

Figure 5. A joint display from a mixed methods intervention design organized by 4 categories of patient treatment benefits.

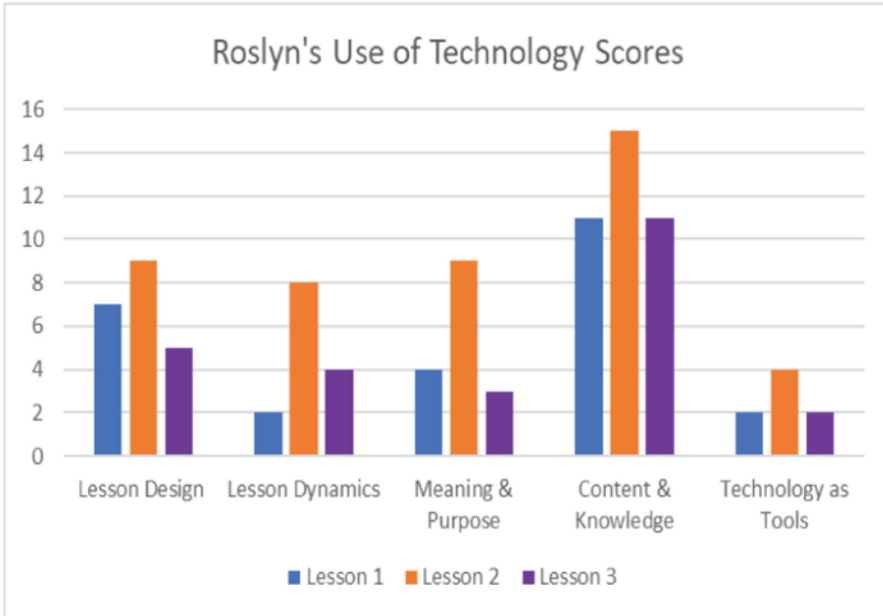
This figure was uploaded by [John W Creswell](#)

Content may be subject to copyright.

Teacher-Centered Case

Roslyn

Quantitative Observation Results on Use of Technology based on FIT:COM Instrument

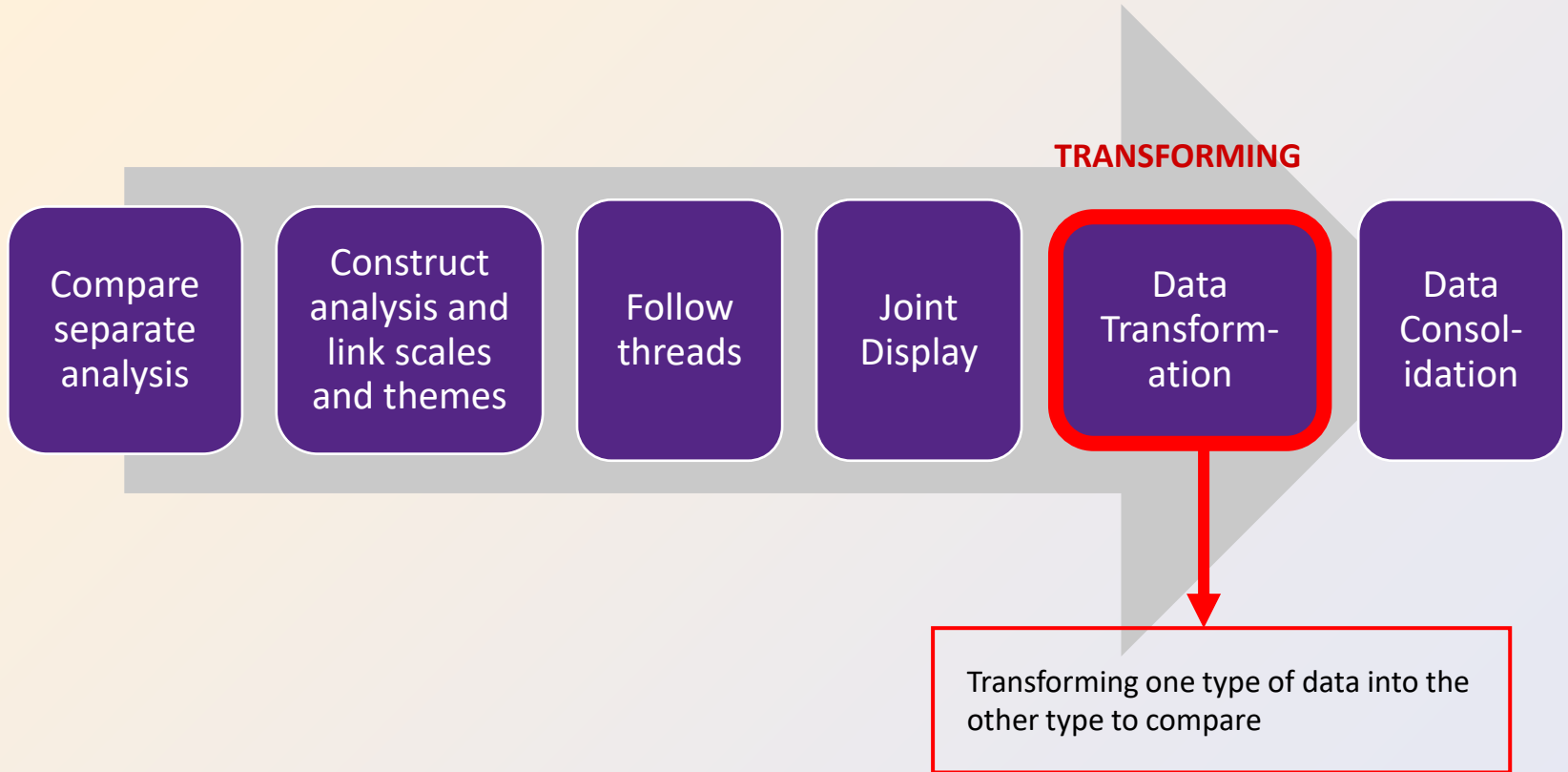


Qualitative Field Notes from Lesson Observations (Typical Pattern Exemplar)

Technology as Tools: The lesson did not include students' use of technology for themselves. At no point in the lesson were the students allowed to touch the devices. The PowerPoint presented notes on fire safety that students read and copied to their notebooks (Field Notes of Lesson Observation 3, Roslyn).

Lesson Design: The video showed an animated demonstration of how the engine of an automobile works. The video aided in proving an explanation of how engines work. The students answered primarily recall level questions after watching the video (Field Notes of Lesson Observation 2, Roslyn).

Techniques for Merging QUAL and QUAN



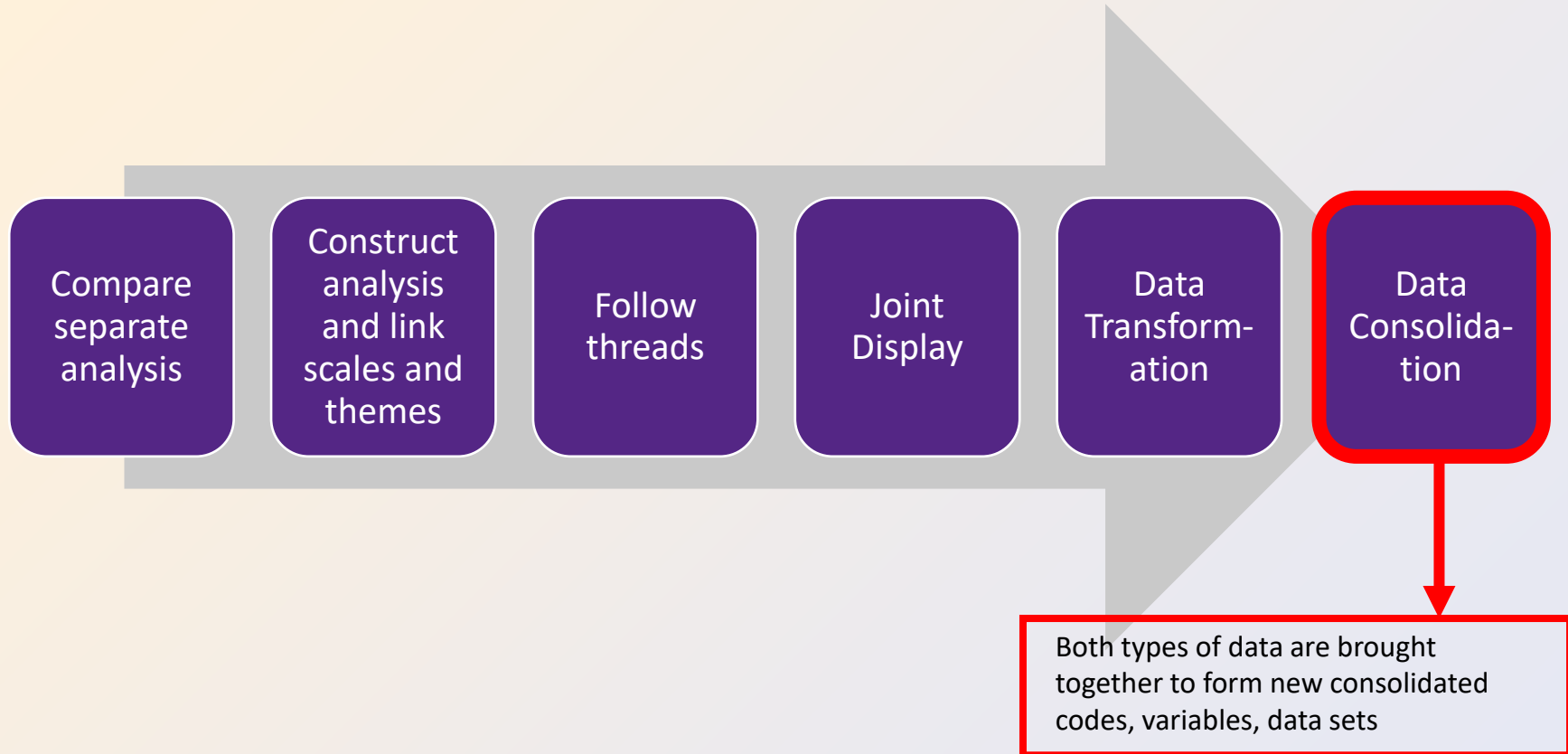
Examples of Transformation Analysis

Creating narrative descriptions of factors from quantitative data to compare with qualitative themes

Create counts of qualitative data (e.g., content analysis) to compare with quantitative

Conduct qualitative assessments of housing fidelity, create categorical variables and include in regression analyses

Techniques for Merging QUAL and QUAN



Drawing Inferences

Inferences = conclusions and interpretations from the data

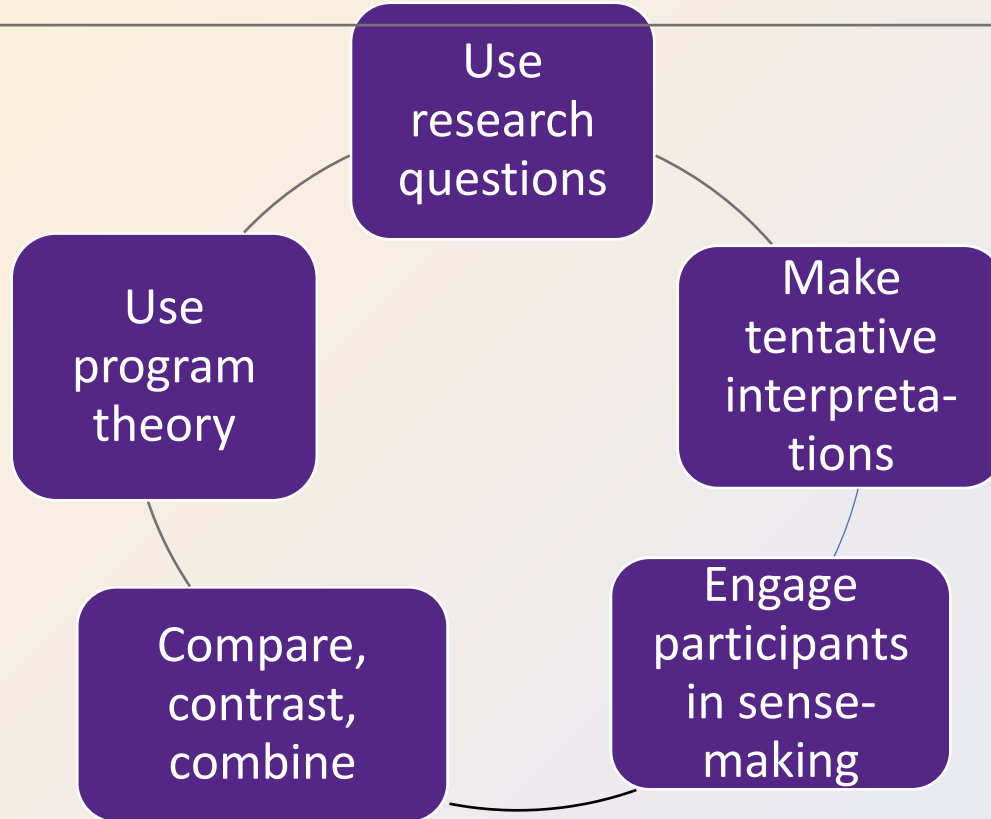


Need to link or integrate data on outcomes to have an enhanced understanding of findings



Drawing inferences is a process – examining the results, making a conclusion, evaluating the conclusion, revising if needed, and moving to the next set of inferences

Guidelines for Drawing Inferences



Ensuring Validity of Conclusions

Focuses on the concept of validity threats

Test the plausibility of specific threats and try to rule them out

Focus on threats that are most plausible, not all possible threats

Work against confirmation bias:

- Ask others to review findings; look for flaws in logic
- Look for evidence that challenges the conclusions or suggests a threat is possible

Mixed Methods Re-Cap

Intentional Integration

Intentional integration of qualitative and quantitative methods

Complementarity

Combination of methods that are complementary, compensating for weaknesses and capitalizing on strengths

Comprehensive

Tell a more complete story

Credibility

Strengthen the credibility of studies for diverse audiences

References

- Creswell, J. W., & Clark, V. L. P. (2017). *Designing and conducting mixed methods research*. Thousand Oaks: Sage Publications
- Guetterman TC, Fetters MD, Creswell JW. Integrating Quantitative and Qualitative Results in Health Science Mixed Methods Research Through Joint Displays. *Ann Fam Med*. 2015 Nov;13(6):554-61. doi: 10.1370/afm.1865. PMID: 26553895; PMCID: PMC4639381.
- Haynes-Brown, T. K., & Fetters, M. D. (2021). Using Joint Display as an Analytic Process: An Illustration Using Bar Graphs Joint Displays From a Mixed Methods Study of How Beliefs Shape Secondary School Teachers' Use of Technology. *International Journal of Qualitative Methods*, 20. <https://doi.org/10.1177/1609406921993286>
- Jang, E. E., McDougall, D. P, Herbert, M., & Russell, P. (2008). Integrative mixed methods data analytic strategies in research on school success in challenging circumstances. *Journal of Mixed Methods Research*, 2, 221-247. DOI: 10.1177/1558689808315323
- Teddlie, C., & Tashakkori, A. (2009) *Foundations of Mixed-methods Research: Integrating Quantitative and Qualitative Approaches in the Behavioral Science*. Thousand Oaks: Sage Publications.